

AURA TERMINAL

Architecture & Engineering Documentary

LEAD ENGINEER: ARNAV VITHLANI • SYSTEM VERSION: 1.0.0 • JULY 2026

This document serves as the official post-mortem and architectural teardown of the AURA Market Intelligence project. What began as a standard academic requirement for the Corizo ML minor project evolved into a production-grade, full-stack quantitative finance application. This documentary chronicles the engineering decisions, algorithmic pivots, and UI/UX refinements that defined the build.

PHASE I: THE DATA INGESTION PARADIGM

The initial concept relied on static CSV files downloaded from Kaggle. However, financial markets are dynamic; a static file is obsolete the moment it is saved. To achieve a real-world edge, the data pipeline was rewritten to utilize the **yfinance REST API**. This upgrade allowed the terminal to fetch live historical data on command. Rather than analyzing a dead dataset, the engine now breathes live market telemetry, seamlessly switching between Top 10 Global Titans and Top 40 Indian Heavyweights.

PHASE II: THE FEATURE FACTORY

Raw closing prices provide insufficient signal for machine learning algorithms. To give the model "vision," a mathematical Feature Factory was engineered to calculate Wall Street-grade indicators:

- **RSI (Relative Strength Index):** To detect overbought or oversold momentum.
- **Bollinger Bands:** To establish upper and lower volatility corridors.
- **Moving Averages (10-day & 50-day):** To smooth out noise and expose underlying trendlines.
- **MACD:** To measure the divergence of short and long-term moving averages.

PHASE III: THE ALGORITHMIC PIVOT (THE EXTRAPOLATION TRAP)

A critical engineering failure occurred during initial model testing with *Random Forest Regression*. Tree-based models map historical averages but cannot extrapolate beyond the

maximum values seen in their training data. When Apple Inc. reached all-time highs, the Random Forest model flatlined, falsely predicting a catastrophic drop.

The Fix: The engine block was swapped from Random Forest to **Ridge Regression**. As a regularized linear model, Ridge accurately tracked the upward momentum vector alongside the technical indicators, dropping the RMSE (Root Mean Square Error) to single digits and closely hugging the actual spot price trajectory. Furthermore, chronological data integrity was enforced (*shuffle=False*) to prevent future data from leaking into the training set.

PHASE IV: THE PRESENTATION LAYER

To transcend the confines of a Jupyter Notebook, the frontend was constructed using **Streamlit** and **Plotly**. The design directive was strict: a minimalist, "old money" aesthetic favoring deep charcoal backgrounds, elegant typography, and subtle emerald/gold data visualization.

Simple line graphs were abandoned in favor of professional Japanese Candlestick charts (OHLC). A dynamic timeframe selector was introduced, allowing users to zoom seamlessly from a 1-Month micro view to an All-Time macroeconomic perspective.

PHASE V: REFINEMENT & QUALITY ASSURANCE

As the system approached deployment readiness, several crucial bugs were identified and eradicated during final QA:

- **Dynamic Localization:** Hardcoded currency symbols were replaced with a localized variable, ensuring Indian equities displayed in Rupees (₹) and US equities in Dollars (\$).
- **UI Symmetry:** A CSS flexbox stretching bug caused the metrics row to misalign. Column ratios were strictly enforced to *[1, 1, 1, 1]* and labels were streamlined for absolute symmetry.
- **Adaptive Theming:** Hardcoded hex colors caused visual collapse when the user switched to Light Mode. The CSS was refactored to utilize Streamlit's native CSS variables (e.g., *var(--secondary-background-color)*), making the UI fully responsive to system theme changes.

PHASE VI: THE AI EXECUTIVE AGENT

To bridge the gap between predictive modeling and generative AI, an Algorithmic Analyst Agent was integrated. Instead of requiring external LLM API keys, a highly sophisticated Python logic engine was coded. It parses the latest RSI, MACD, and Bollinger Band metrics and dynamically generates a bespoke, human-readable market briefing. This provides instant qualitative context to the quantitative data.

